

Coordination of Automated Vehicles at a Traffic-Lightless Intersection

M.A.S. Kamal, J. Imura, A. Ohata, T. Hayakawa and K. Aihara

Abstract—This paper presents a coordination scheme of automated vehicles at an intersection without using any traffic lights under a connected vehicles environment. Vehicles approaching the intersection from all sections are globally coordinated, by considering their states all together in a model predictive control framework, for their safe and rapid crossing. The optimal trajectories of the vehicles are computed based on avoidance of the cross-collision risks around the intersection. Compared with the other methods, the scheme enhances safety by generating the trajectories of vehicles far from each other with respect to the cross-collision points and enables turning movements under constrained velocity without using any auxiliary lanes. The proposed coordination scheme is evaluated through numerical simulation for a simple test intersection consisting of four single-lane approaches. Observations under different traffic flow conditions reveal that the proposed scheme significantly improves performance compared with the traditional traffic-light-based intersection control.

I. INTRODUCTION

Traffic congestion in urban road networks wastes a massive amount of time, fuel and money, and the overall cost due to traffic congestion is increasing every year [1]. Traffic congestion often occurs when traffic density becomes higher than the capacity of roads. Efficient use of the existing roads by innovative traffic management and control is a remaining option for the cities where further construction and expansion of roads are difficult. In the traditional paradigm traffic flows at intersections are regulated by traffic lights or signs that restrict the capacity of the intersections and increase inconveniences of frequent stops and idling. The intersections with light traffic are usually operated by yield and stop signs. A vehicle at a yield sign only has to stop if it is necessary to let a vehicle on another approach have the right-of-way first. In contrast, vehicles at a stop sign must come to a complete stop before crossing, even if no other vehicle is present. Busy intersections are usually regulated by green and red lights that eventually increase the stop delay.

Among the various solutions to relieve traffic congestion, traffic-light or signal control is commonly considered as the most effective method, and various strategies for urban traffic management have been developed [2], [3]. These strategies can only partially improve the traffic flows if all approaches to the intersection are not equally congested, and they cannot eliminate the stop delay of vehicles at

intersections. Considering increasing trends of vehicles and sustainability of the road transportation, a break through in intersection control paradigm, which may eliminate the necessity of stop and increase the capacity of intersections, is highly expected.

Semi-automated vehicles are already emerging in the market, and it is expected that fully automated vehicles will be widely used in the near future considering the advancement in sensor technology and adoption of communication technology to create a fully connected vehicles environment. Connected vehicles provides a two-way wireless communication environment enabling vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communications [4]. Using V2V communication, cooperative adaptive cruise control (CACC) systems can safely drive vehicles with very short headway by forming platoons to improve traffic flow capacity of a road [5]. The concept of closely following in CACC can be extended to develop a new intersection control paradigm in which vehicle can cross an intersection keeping marginal gaps with other crossing vehicles without using any traffic signal. Such an intersection control paradigm would unleash the full potential of automated vehicles in mostly eliminating stop delay, reducing travel time and increasing capacity of the intersections.

A few studies on innovative intersection management have been found in the literature that attempts to control vehicles without using traffic signals. Raravi et al. [6] proposed an automatic merge control system for intelligent vehicles under a cooperative vehicle-infrastructure environment that ensures safe vehicle maneuver at road intersections. By formulating an optimization problem with constraints to guarantee safety, the optimal maneuvers for merging vehicles were obtained by minimizing the maximum of driving time to intersection for every vehicle coming from two conflicting approaches. Uno et al. [7] also studied the merge control application based on V2V communication under the concept of virtual vehicles that is used to map vehicles on one lane onto the other lane for ensuring safe distance criteria. These merge control systems are not suitable for intersections that needs to be managed for complex movements of multiple vehicles to and from various sections.

Dresner et al. [8] presented an idea of autonomous intersection management (AIM) as an alternative to the traditional traffic signal control mechanism. In AIM, vehicles and intersections are treated as autonomous agents in a multiagent system. The intersection is divided into a number of cells, and an intersection manager program coordinates the reservation requests of temporal cell occupancies from every vehicle, and offers right-of-way for each vehicle for ensuring

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safe crossing. However, this method does not coordinate the vehicles globally for their optimal flow, and therefore stop delay cannot be avoided sometimes.

Recently a cooperative vehicle intersection control (CVIC) system has been proposed that enables cooperation between automated vehicles and infrastructure for effective intersection operations [9]. CVIC algorithm is based on minimizing the overlaps of projected trajectories of conflicting vehicles in the intersection area. More specifically, this system simply tries to avoid presence of any pair of conflicting vehicles in the intersection area. However, presence of two conflicting vehicles just at the outside of intersection is considered safe, although such condition may cause collision at some corner in the intersection depending on the lengths of the vehicles. Thus, this system keeps the optimization problem simple without including any constraints for ensuring cross-collision avoidance, and as a result, sometimes it fails to find feasible trajectories of the vehicles. Hence, it requires a system recovery control algorithm to handle such malfunctions. Furthermore, CVIC proposed in [9] was applied only for the traffic that move forward without any turning at the intersection.

This paper proposes a coordination scheme of automated vehicles at an intersection, without using any traffic lights under a connected vehicles environment, that overcomes the limitations of CVIC. The novelty of the proposed scheme is the global coordination, by considering the states all vehicles together, based on avoidance of their projected cross-collision risks around the intersection. Moreover, introduction of relevant constraints ensures the scheme free from any collisions and enables to manage turning movements of the vehicles under a safe velocity limit without using any auxiliary lanes. An intersection coordination unit (ICU), which is assumed to be installed at the intersection, uses two-way communication to receive basic driving information from the approaching vehicles, e.g., current position, velocity and destination at the next intersection, and send guidance to them after computing their control inputs. In this computation, considering states of all vehicles a constrained nonlinear optimization problem that includes a risk function is solved in a model predictive control framework in order to let them cross the intersection safely without much drop in their velocities. The risk function quantifies the risk of a projected collision in terms of distances to the center of a cross-collision point of any pair of conflicting vehicles in the intersection area. By minimizing the total quantified risks, the trajectories of the vehicles are kept far from each other with respect to the cross-collision points. Thus the scheme integrates fully automated vehicles in a fully automated intersection to improve traffic operations. The proposed scheme has been evaluated through numerical simulation and its performance is compared with the traditional signal-based control. In various traffic flow conditions it is observed that the stop delay of vehicles at the intersection is almost eliminated, and flows of vehicles and the capacity of the intersection are improved significantly.

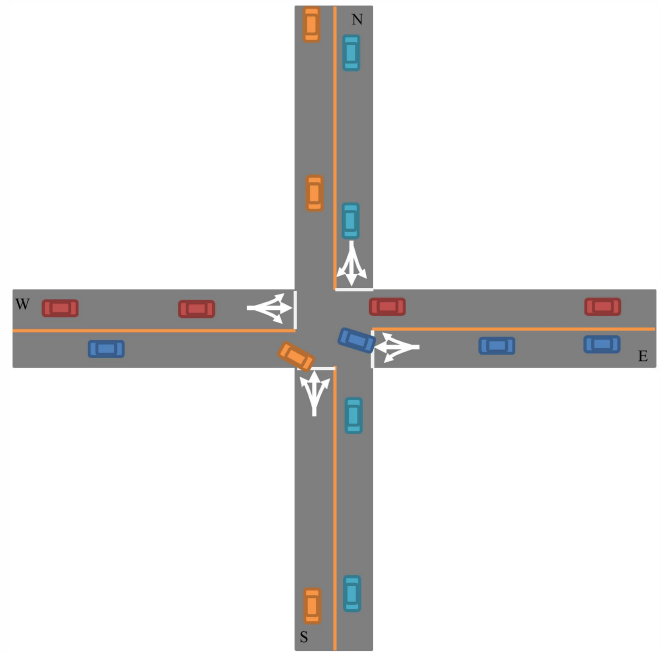


Fig. 1. An intersection consists of four single-lane approach sections with flat terrain in which vehicles are allowed to turn without using any auxiliary lanes.

II. COORDINATION OF VEHICLES AT INTERSECTION

A. Problem Statement and Assumptions

The coordination of automated vehicles in a small intersection is considered in this paper. Specifically, Fig. 1 shows an intersection in which traffic keeps to the left side of the road according to the convention in Japan. Each of the four approach sections and their extension beyond the intersection has a single lane with flat terrain, and vehicles are allowed to make left or right turn without using any auxiliary lanes. It is assumed that all automated identical vehicles crossing the intersection can be manipulated by an intersection coordination unit (ICU) equipped at the intersection through two-way communication using standard communication protocol as defined in Wireless Access in Vehicular Environments Short Range Communications standards [4], [10]. It is assumed that under such connected vehicles environment every vehicle transmits its basic driving information including current position, velocity and destination at the next intersection, and the ICU transmits the corresponding guidance to each vehicle with negligible delay. Only the longitudinal motion of the vehicles are controlled, assuming the vehicles perfectly follow the lateral trajectories. The maximum velocity of longitudinal motion when a vehicle turns is constrained for safety. Vehicles approach the intersection using their independent driving systems, and only a few leading vehicles on each section need to be globally coordinated considering the trajectories of the whole vehicles in question. Once a vehicle crosses the intersection, its control is transferred to its independent driving system, and a new set of vehicles are optimized for crossing the intersection in a successive approach.

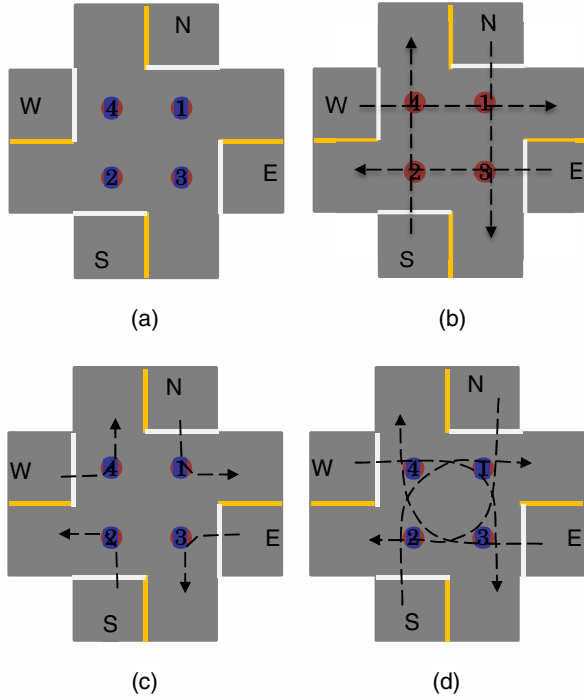


Fig. 2. (a) Cross-collision points in the intersection area; and assumed standard path of vehicles from four approaches for crossing the intersection: (b) going straight, (c) turning left, and (d) turning right.

In order to formulate a vehicle coordination scheme, some variables are defined as follows.

- $L = \{N, S, E, W\}$ set of approach lanes. Since every section has single lane, depending on the context ‘section’ and ‘lane’ are interchangeably used to denote the same.
- x_n^l the remaining distance from a vehicle to the beginning of an intersection of vehicle n on lane $l \in L$. Note that at lane l vehicle n is closer to the intersection than vehicle $n+1$, i.e., $x_n^l < x_{n+1}^l$.
- v_n^l the velocity of a vehicle.
- u_n^l control input of the vehicle in terms of acceleration. If its magnitude is negative then it implies deceleration.
- t time step counter.
- τ duration of the time step.
- $d_n^l \in \{0, 1, 2\}$ vehicle’s destination at the intersection as straight, left or right, respectively.

The discrete-time state equations related to the longitudinal motion of a vehicle approaching the intersection are given by

$$\begin{aligned} x_n^l(t+1) &= x_n^l(t) - v_n^l(t)\tau - \frac{1}{2}u_n^l(t)\tau^2, \\ v_n^l(t+1) &= v_n^l(t) + u_n^l(t)\tau. \end{aligned} \quad (1)$$

B. Cross-Collision Points and Risk Function

A new concept of a risk function that quantifies a projected risk of a cross-collision of a pair of vehicles in the intersection area is described here. Fig. 2(a) shows the four red points numbered 1 to 4 as the crossing points of vehicles from the four sections, and they are named as *cross-collision points (CCP)*. The position of these CCPs $p = 1, \dots, 4$ from

the beginning of the intersection viewing from section $l \in L$ are identified as z_p^l . For example, vehicles from both the north and the west sections move near CCP 1. Note that $|z_1^N| < |z_1^W|$, since CCP 1 is closer to the north section (the line marked by the while line) than the west section. Therefore, there is a risk of collision if vehicles from the north and the west sections approach CCP 1 at almost the same time.

Figs. 2(b), (c) and (d) show the assumed standard paths of vehicles that are going straight, turning left and turning right, respectively, from the four sections. The vehicles that turn left at the intersection move near only one CCP, whereas vehicles that go straight or turn right move near three CCPs. A binary variable θ_{ln}^p is introduced to map the destination d_n^l of a vehicle to the CCPs as illustrated in these figures. Specifically, $\theta_{ln}^p = 1$ defines the case where vehicle (ln) passes over CCP p . A risk function $f_{ln,km}^p(t)$ is defined to quantitatively indicate whether the vehicle pair (ln) and (km) poses a potential risk of collision at CCP p at time t given by

$$f_{ln,km}^p(t) = H\delta_{ln,km}^p e^{-(\alpha_l^n(x_n^l(t)-z_p^l)^2 + \alpha_k^m(x_k^m(t)-z_p^k)^2)}, \quad (2)$$

where H, α_l^n and α_k^m are constants, and

$$\delta_{ln,km}^p = \begin{cases} \theta_{ln}^p \theta_{km}^p, & \text{if } k \neq l; \\ 0, & \text{otherwise.} \end{cases}$$

If $H\delta_{ln,km}^p = 1$ the risk function becomes simply a two-dimensional Gaussian function, and α_k^m and α_l^n are related to its shape (or the area under high risk). By adjusting α_k^m and α_l^n , area of risk for small to large vehicles can be defined. Risk function (2) implies a high risk of collision, $f_{ln,km}^p(t) \rightarrow H$, if both vehicles are at the center of CCP p and a negligible risk of collision, $f_{ln,km}^p(t) \rightarrow 0$, if one or both of them are far from CCP p . For safe intersection crossing it is necessary to ensure that at any CCPs the risk of any pair of vehicles must be close to zero.

III. MODEL PREDICTIVE CONTROL

Considering the state equation of vehicles (1) and risk function (2) related to the CCPs in the intersection, in this section a vehicle coordination scheme in a model predictive control framework is formulated. In the model predictive control framework, at each time interval a constrained non-linear optimization problem is solved over a finite horizon to derive the optimal control sequences for vehicles. For the safe flow of vehicles both on the sections and intersections, several constraints are considered. Specifically, linear inequalities

$$v_n^l \leq v_{max}, \quad (3a)$$

$$v_n^l \geq v_{min}, \quad (3b)$$

$$u_n^l \leq u_{max}, \quad (3c)$$

$$u_n^l \geq u_{min}, \quad (3d)$$

are defined that are related to the velocity and acceleration limits of a vehicle. Although maximum possible acceleration depends on the current velocity of the vehicle, for simplicity

in this study constant magnitudes are considered as the limits. The minimum separation between two consecutive vehicles on the same section l is linearly constrained by

$$x_{n+1}^l - x_n^l \geq r_{min}. \quad (4)$$

Some vehicles may turn left or right at the intersection, and their velocities at the intersection must be reduced to some level for safety as per standard traffic rules. Considering the fact, a nonlinear constraint for the vehicle with $d_n^l \neq 0$ is given by

$$v_n^l \leq \left(V_a(1 - e^{-\beta(x_n^l + b)^2} + V_b) \right), \quad (5)$$

where β is a constant, b is the distance between the beginning and the center of the intersection, V_a and V_b are nonnegative constants. The nonlinear constraint (5) implies that at $x_n^l + b = 0$ (at the center of the intersection) the constraint becomes $v_n^l \leq V_b$, and at $|x_n^l + b| \gg 0$ (the vehicle is very far from the intersection) the constraint becomes $v_n^l \leq V_a + V_b$. For ensuring no collision between two vehicles coming from different sections (l and k) at CCP p in the intersection, a nonlinear constraint is defined as

$$(x_n^l - z_p^l)^2 + (x_m^k - z_p^k)^2 \geq R_{min}^2, \quad (6)$$

where R_{min} is the minimum separation required to avoid any collision of two vehicles at the CCPs.

Considering the computation burden, only M number of vehicles from each section and total of $4M$ vehicles are globally coordinated for safe intersection crossing. Subject to the given current positions and velocities of vehicles, their state equations (1) and constraints (3)–(6), a performance index including a risk function (2) for a finite horizon of T steps is considered as to minimize the cost function

$$\begin{aligned} J = & \sum_{t=0}^{T-1} \left(w_v \sum_{l \in L} \sum_{n=1}^M \left(v_n^l(t+1) - v_d \right)^2 \right. \\ & + w_u \sum_{l \in L} \sum_{n=1}^M \left(u_n^l(t) \right)^2 \\ & \left. + \sum_{p=1}^4 \sum_{l \in L} \sum_{n=1}^M \sum_{k \in L} \sum_{m=1}^M f_{ln,km}^p(t+1) \right), \end{aligned} \quad (7)$$

where v_d is the desired velocity, w_v and w_u are weight coefficients. There are three cost terms in the above performance index. The first term denotes the cost related to velocity deviation from a desired value v_d . The second term denotes the cost of acceleration. Minimizing these two terms means comfortable and smooth flow of vehicles at a standard velocity. The third term denotes the cost related to risk of collisions as defined in the risk function (2) which sums up quantified risks at the four CCPs for all possible pairs of vehicles considering their predicted trajectories in the horizon. Similar to a barrier function, this term is included in the performance index to distinguish the infeasible space of high risk, so that the optimal solution is sought outside infeasible region.

At each discrete time step t , ICU receives information of current position, velocity and destination of the vehicles and computes their control inputs u_n^l for the entire horizon by

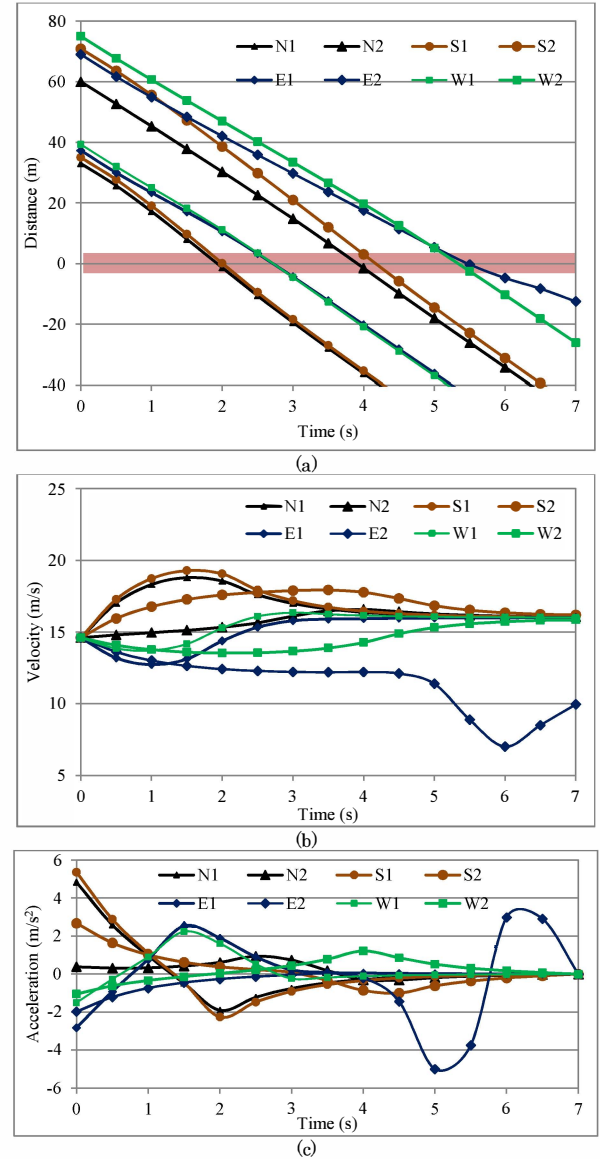


Fig. 3. Results of optimization at one step for a test case, (a) distance from the center of the intersection (b) velocities and (c) acceleration of vehicles approaching the intersection from the four sections. Turning velocity of vehicle E2 is constrained, and the other vehicles move straight.

solving the above nonlinear constraint optimization problem. However, only the control inputs corresponding to the current time are applied to each vehicle. It is assumed that each vehicle has a local controller installed that perfectly generates and executes the actual engine torque considering the vehicle-engine dynamics for realizing the desired acceleration received from the ICU. At the subsequent time step the optimization procedure is repeated for newly measured initial conditions and for a new set of vehicles if some vehicles have already crossed the intersection.

IV. NUMERICAL SIMULATION

The proposed coordination of vehicles is numerically simulated for the test intersection shown in Fig. 1. The constrained nonlinear optimization problem in the proposed

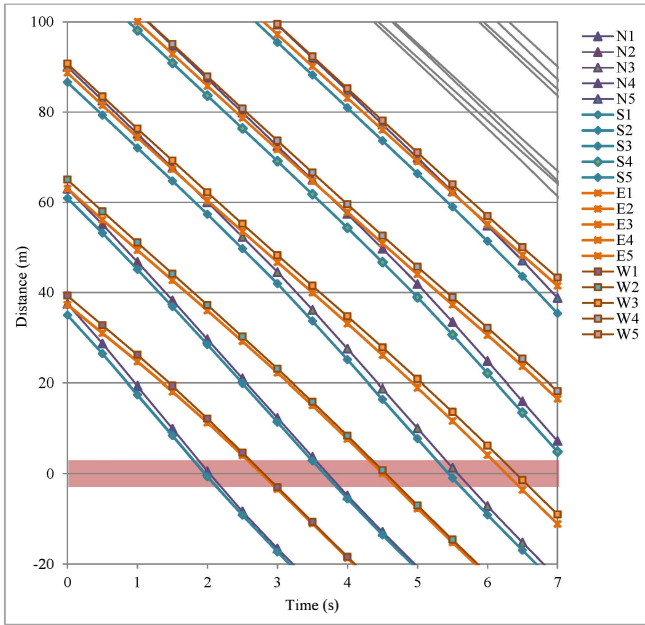


Fig. 4. Trajectories of vehicles crossing the intersection which are sequentially coordinated by the proposed scheme in the model predictive control framework.

model predictive control framework is solved in MATLAB using the optimization toolbox *fmincon*. Only $M = 2$ approaching vehicles which are the closest to the intersection from all four lanes are globally coordinated at a time, which means trajectories of 8 vehicles are optimally controlled by ICU based on the state of the whole vehicles in question. The discrete time step is chosen as $\tau = 0.5$ s, and the prediction horizon as $T = 14$ steps. The parameters of the linear constraints are chosen as $v_{max} = 23$ m/s, $v_{min} = 0$, $u_{max} = 5$ m/s², $u_{min} = -6$ m/s², $r_{min} = 8$ m and $R_{min} = 10$ m. The constraint related to the turning velocity (5) is set with $V_a = 14$, $V_b = 7$, $\beta = 0.003$ and $b = 3$. The parameters of the risk function (2) is chosen as $H = 1000$ and $\alpha_n^l = 0.005$ for all identical vehicles. The performance index is set with $v_d = 16.67$ m/s (60 km/h), $w_v = 2.0$ and $w_u = 5$.

To comprehend the functionality of the proposed scheme, the results of the optimization for a certain initial states of vehicles are observed. All the vehicles cross the intersection straight except vehicle E2, which turns left under a constrained velocity at the intersection. Fig. 3 shows the results of optimization as predicted trajectories of the vehicles, corresponding velocities and control inputs of the vehicles. The vertical axis in Fig. 3 (a) shows the remaining distance from the current vehicle position to the center of the intersection, and the shaded area indicates the intersection. Although the leading vehicles on the four lanes have almost the same velocities and distances from the intersection, their future trajectories are optimized in such a way that the conflicting vehicles do not pass over intersection (the shaded area) at the same time.

The actual coordination of vehicles is implemented in the MPC framework for continuous operation of the intersection.

For its evaluation, a realistic traffic environment is created inspired by the AIMSUN simulator [11], where the vehicles stochastically arrive at the intersection. Each of these vehicles are controlled independently using microscopic car following model with a maximum velocity of about 52 km/h. Fig. 4 shows the trajectories of a few vehicles on each section that cross the intersection straight away (without turning). These vehicles are coordinated sequentially to cross the intersection in such a way that the conflicting trajectories are widely separated from their corresponding CCPs.

Fig. 5 shows the overall performance of the coordinated vehicles in terms of their average velocities and normalized acceleration. In this case, input traffic rates are equally set for all sections and all vehicles go straight without turning at the intersection. The horizontal axis shows the average traffic volume per lane as observed in numerical simulation for a range up to about 1800 veh/h. Each marker in the graphs shows the observed values and the curves represent the trends in polynomial approximation. It should be noted that in the traditional signal-based system as observed in microscopic traffic flow simulation, at most about 900 veh/h (saturation flow rate 1800 veh/h and 50% Green time) can cross the intersection. For low traffic volume, up to 900 veh/h, the vehicles are able to cross the intersection without dropping their velocities and with little acceleration (average of normalized values). The proposed scheme is capable of handling the traffic higher than the capacities of traditional signal-based intersection. At higher traffic, the average velocity remains above 50 km/h, but the average acceleration increases considerably, which is due to increasing number of conflicting vehicles that require variation in their velocities. The most significant achievement is the elimination of stop delay at the intersection. Although the vehicles arrive with a maximum velocity of about 52 km/h, the desired crossing velocities in the proposed scheme is set at a higher values of 60 km/h. Therefore, when there is no turning vehicles whose velocity is constrained at some low value, the vehicles cross the intersection more rapidly than their arrival rate, and consequently, the capacity of the intersection increases by reducing any chance of bottleneck.

Fig. 6 shows a comparison of approximate capacities of the traditional signal-based and the proposed coordinated intersections for different rates of left turning vehicles, which are observed through simulation. For this comparison, the same traffic is also separately controlled by traditional method using two-signal phase, and setting 50% green time and 90 s cycle time. Since some vehicles turn at the intersection, as per standard rules for safety, they reduce the velocity under a safe level, which also affects flows of the vehicles behind them. Therefore, the capacity drops as the turning rate of vehicles increases. Regardless of the rate of turning, the maximum capacities of the intersection become almost double by the proposed scheme.

In addition to the above observations, the proposed scheme has also been tested for operating the traffic that includes both the left and the right turning vehicles with constrained velocity at intersection. Note that using two-signal phase

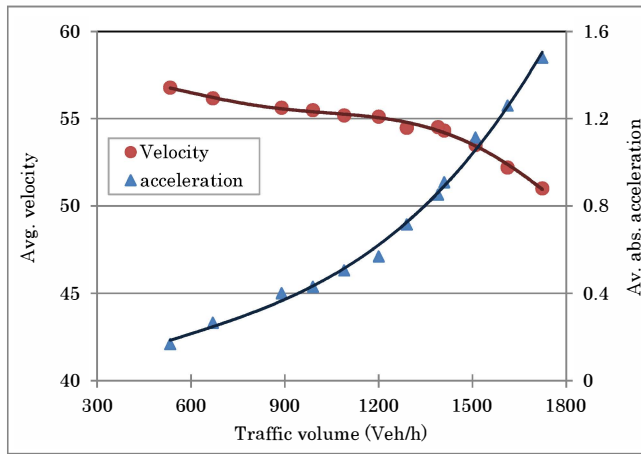


Fig. 5. Average velocities and average normalized acceleration of vehicles which are coordinated by the proposed scheme for crossing the intersection.

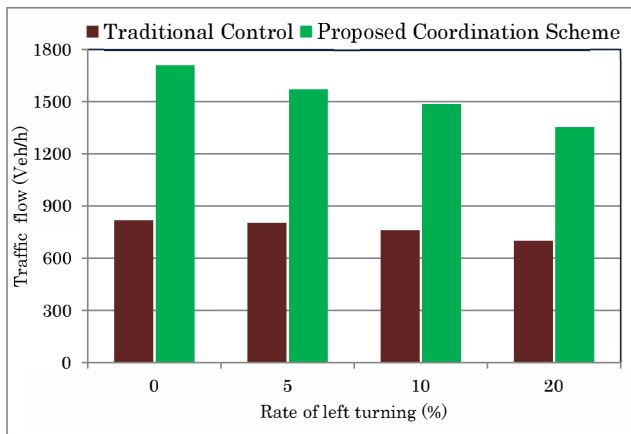


Fig. 6. Maximum possible capacities of the intersection for different rate of left turning vehicles observed in numerical simulation: comparison of the proposed scheme with the traditional signal-based intersection control.

traditional intersection cannot allow right turning of vehicles at all. It is found that the proposed scheme can handle the mixed traffic in the same way although there is no auxiliary lane for right turning. However, right turning vehicles take longer time to pass over the intersection than left turning vehicles and usually cause conflicts with vehicles from all other lanes. Therefore, the maximum capacity of the intersection also drops further.

The results regarding the maximum capacity is observed only to quantify the potential upper ceiling limits of the coordinated intersection. Considering the possible anxiety and discomfort of drivers due to very close passing, such maximum limits may not be used in real implementation at all. Still it can bring huge benefits as the system can simply replace the traditional intersection for handling the same traffic. Since the proposed scheme improves travel time, reduces acceleration and almost eliminates stop delay as observed from simulation, it is obvious that implementing such schemes would significantly reduce the energy consumption and CO₂ emissions in the road transportation.

V. CONCLUSIONS AND FUTURE WORK

A vehicle coordination scheme for a traffic-lightless intersection has been proposed under a connected vehicle environment. Vehicles approach the intersection from all sections are globally coordinated in order to allow them to cross the intersection safely without much dropping the velocities. The control inputs of the vehicles are optimized in a model predictive control framework by minimizing the risks of projected cross-collisions of the vehicles in the intersection area and considering various constraints related to their safe crossing or turning at the intersection. The proposed vehicle coordination scheme has been evaluated in numerical simulation and its performance is compared with traditional signal-based intersection control. Through global coordination the stop delay of vehicles at the intersection is almost eliminated so that traffic flows are improved and consequently the capacity of the network is increased.

The future research will deal with further evaluation of the scheme for multiple intersections along the roads or in a network and comparison of other achievements, such as fuel consumptions, CO₂ emissions, for more realistic cases.

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